

UCF Physics AST4762/5765: Introduction to Numerical Computing

Keeping a worklog

The purpose of a lab log is to make a record of what you did. Mostly, this is useful to you and your supervisor. Logs are also the main items scrutinized by Federal research audits. In commercial research, they establish priority for patents and other forms of intellectual property (who made the key discovery first?). If you are in such a field, you will likely have detailed procedures to follow, e.g., handing in paper log pages each day to your employer's attorney, for notarization and archiving.

Our logs are online, so they are searchable. You can date-match notes to email (don't insert the email in the log, just keep it and don't delete it). The log will save you time in the future. The log has dated entries for all work sessions. Record summaries of what you hear and do, whom you meet, and what was said and agreed in meetings. Keep to-do lists there. Record links to online resources, snippets of code, pointers to screen shots and plots, etc.

This enables you to...

- Go back later and restart work
- Figure out what went wrong and fix it
- Manage a large number of trials
- Recall your thinking
- Prove/report your activities to others
- Track time

The most important thing is that you **never change anything above the current entry!**

To make the log:

1. Use a plain text editor. Not Word (Windows) or Pages (Mac) or similar programs. For Windows, try Start->Windows Accessories->Notepad. On Mac, launch Finder->Applications->TextEdit, then immediately, TextEdit->Format->Make Plain Text. gedit is a good choice under Linux.
2. Name the file `0-ast4762-yourname.log` (in general, if you keep log for another project, say e.g., your research lab, you can name it `0-researchlab-yourname.log`). Don't forget to change *yourname* with your name! You should keep this in the top level of the `ast4762/` folder (or any other folder you work in for other projects). See the `homework_format` pdf for a schematic.

At the top of the file you should start with stating the purpose of this log. For example:

```
Jane Doe  
Log for AST 4762, Fall 2023
```

Now we will start filling your log in.

1. Start every new entry at the bottom of the log (remember, we **never** change anything before the new entry)
2. Start by indicating that this is a new entry by letting the reader know when you start as :

```
*****  IN : Thursday Aug 24 12:45 pm ET 2023
```

3. For readability leave a blank line
4. Now you can add information on the log about what you are doing. Make sure you are descriptive. Keep notes about where you work, what you do and how you do it. For example:

```
Went into the ast4762/homework/ folder using the
Finder. Right clicked and made a new folder for
homework 3. I named it hw3_jdoe/. I then used the
Anaconda Navigator to launch Jupyter notebook and
created a new notebook in ast4762/homework/hw3_jdoe/.
Went on the top left corner of the notebook and
changed the name of the notebook from Untitled.ipynb
to hw3_jdoe.ipynb.
```

5. It is good practice to keep information about results you get from your project. Also, don't just report a result, *think critically* about it. If you realize something doesn't make sense and you note it, you will get partial points.

For the homework for example, if we ask you to calculate the speed of an object and you calculate it to be $3.5e9$ m/s your log entry could look like this:

```
Started Problem 2. Called the function speed() and
calculated the speed of the car using the given
distance and time. The function returned a speed of
 $3.5e9$ m/s. This is obviously wrong because it is larger
than the speed of light. I cannot find the bug at this
time. Will try to get back to it later.
```

```
TO DO: find bug;
        continue with rest of homework
```

6. Once you are done with your current working session (we all need brakes, and we all have multiple things we need to do in a day) don't forget to log out. To do this, leave a blank line after the last entry and then type OUT and give information about when you stop working on the project:

***** OUT : Thursday Aug 24 1:15 pm ET 2023

The row of stars and the exact form of the IN/OUT date lines allow you to use a program like `grep` on your log to print just these lines, and the dates will line up nicely. This is useful when you fill in your timecard or otherwise need to track the time spent on your project. The row of stars makes it easy to search for date entries, either visually or with your editor's search function. Do not put a row of 5 or more stars anywhere else in your log!

Your entire log so far would look like this:

Jane Doe

Log for AST 4762, Fall 2023

***** IN : Thursday Aug 24 12:45 pm ET 2023

Went into the `ast4762/homework/` folder using the Finder. Right clicked and made a new folder for homework 3. I named it `hw3_jdoe/`. I then used the Anaconda Navigator to launch Jupyter notebook and created a new notebook in `ast4762/homework/hw3_jdoe/`. Went on the top left corner of the notebook and changed the name of the notebook from `Untitled.ipynb` to `hw3_jdoe.ipynb`.

Started Problem 2. Called the function `speed()` and calculated the speed of the car using the given distance and time. The function returned a speed of `3.5e9m/s`. This is obviously wrong because it is larger than the speed of light. I cannot find the bug at this time. Will try to get back to it later.

TO DO: find bug;
 continue with rest of homework

***** OUT : Thursday Aug 24 1:15 pm ET 2023

7. Never change anything above the current date line. This can get you fired! In cases of Federally funded research, it is illegal and can wind you up in jail or with a heavy fine for you or your employer for falsifying research. Don't do it! If anyone ever asks you to, report them to your supervisor. If your supervisor asked

you to, tell their boss. Falsifying research is no joke. We never “clean up” our logs. If you need to correct errors:

- I) For task lists, just copy the old list to the current date and mark tasks done or remove them. As your task list gets done, this helps to document when you added and completed the tasks, or how you changed them.
- II) To fix an error: Never edit a prior entry! Instead, record in the current entry a row of XXXXX and the date of the entry you are fixing, and state the change.

The value of a log to you today is modest. You likely remember much more detail about your recent work than is recorded in your log. However, its value rises as memory fades. As you build a body of work, people will ask you about work done years or even decades ago, and you will be able to tell them exactly what you did, when, why, and how. This can earn many benefits, such as co-authoring papers by locating and sharing your old data or model runs, saving time hunting for things, and moving guesswork into the realm of the factual. If you become famous, historians and biographers will love your logs! By never changing anything above the current date line, you can know, swearing under oath in court, if necessary, what you did, when you did it, why, and how.